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Spatial Branching Processes, Random Snakes and Partial Differential Equations

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Foreword

In these lectures, we give an account of certain recent developments of the theory of spatial branching processes. These developments lead to several fascinating probabilistic objects, which combine spatial motion with a continuous branching phenomenon and are closely related to certain semilinear partial differential equations.

Our first objective is to give a short self-contained presentation of the measure-valued branching processes called superprocesses, which have been studied extensively in the last twelve years. We then want to specialize to the important class of superprocesses with quadratic branching mechanism and to explain how a concrete and powerful representation of these processes can be given in terms of the path-valued process called the Brownian snake. To understand this representation as well as to apply it, one needs to derive some remarkable properties of branching trees embedded in linear Brownian motion, which are of independent interest. A nice application of these developments is a simple construction of the random measure called ISE, which was proposed by Aldous as a tree-based model for random distribution of mass and seems to play an important role in asymptotics of certain models of statistical mechanics.

We use the Brownian snake approach to investigate connections between superprocesses and partial differential equations. These connections are remarkable in the sense that almost every important probabilistic question corresponds to a significant analytic problem. As Dynkin wrote in one of his first papers in this area, "it seems that both theories can gain from an interplay between probabilistic and analytic methods". A striking example of an application of analytic methods is the description of polar sets, which can be derived from the characterization of removable singularities of the corresponding partial differential equation. In the reverse direction, Wiener's test for the Brownian snake yields a characterization of those domains in which there exists a positive solution of $\Delta u = u^2$ with boundary blow-up. Both these results are presented in Chapter VI.

Although much of this book is devoted to the quadratic case, we explain in the last chapter how the Brownian snake representation can be extended to a general branching mechanism. This extension depends on certain deep connections with the theory of Lévy processes, which would have deserved a more thorough treatment.

Let us emphasize that this work does not give a comprehensive treatment of the theory of superprocesses. Just to name a few missing topics, we do not discuss the martingale problems for superprocesses, which are so important when dealing with regularity properties or constructing more complicated models, and we say nothing about catalytic superprocesses or interacting measure-valued processes. Even in the area of connections between superprocesses and

partial differential equations, we leave aside such important tools as the special Markov property.
On the other hand, we have done our best to give a self-contained presentation and detailed proofs, assuming however some familiarity with Brownian motion and the basic facts of the theory of stochastic processes. Only in the last two chapters, we skip some technical parts of the arguments, but even there we hope that the important ideas will be accessible to the reader.
There is essentially no new result, even if we were able in a few cases to simplify the existing arguments. The bibliographical notes at the end of the book are intended to help the reader find his or her way through the literature. There is no claim for exhaustivity and we apologize in advance for any omission.
I would like to thank all those who attended the lectures, in particular Amine Asselah, Freddy Delbaen, Barbara Gentz, Uwe Schmock, Mario Wüthrich, Martin Zerner, and especially Alain Sznitman for several useful comments and for his kind hospitality at ETH. Mrs Boller did a nice job typing the first version of the manuscript. Finally, I am indebted to Eugene Dynkin for many fruitful discussions about the results presented here.

Paris, February 4, 1999

Frequently Used Notation

$\mathcal{B}(E)$	Borel σ-algebra on E.
$\mathcal{B}_b(E)$	Set of all real-valued bounded measurable functions on E.
$\mathcal{B}_+(E)$	Set of all nonnegative Borel measurable functions on E.
$\mathcal{B}_{b+}(E)$	Set of all nonnegative bounded Borel measurable functions on E.
$C(E)$	Set of all real-valued continuous functions on E.
$C_0(E)$	Set of all real-valued continuous functions with compact support on E.
$C_{b+}(E)$	Set of all nonnegative bounded continuous functions on E.
$C(E,F)$	Set of all continuous functions from E into F.
$C_0^\infty(\mathbb{R}^d)$	Set of all C^∞-functions with compact support on $\mathbb{R}^d$.
$\mathcal{M}_f(E)$	Set of all finite measures on E.
$\mathcal{M}_1(E)$	Set of all probability measures on E.
$\operatorname{supp}\mu$	Topological support of the measure μ.
$\dim A$	Hausdorff dimension of the set $A \subset \mathbb{R}^d$.
$B(x,r)$	Open ball of radius r centered at x.
$\bar{B}(x,r)$	Closed ball of radius r centered at x.
$\bar{A}$	Closure of the set A.

$\langle \mu, f\rangle = \int f\,d\mu \qquad$ for $f \in \mathcal{B}_+(E),\ \mu \in \mathcal{M}_f(E)$.

$\operatorname{dist}(x,A) = \inf\{|y-x|, y \in A\}, \qquad x \in \mathbb{R}^d,\ A \subset \mathbb{R}^d$.

$$p_t(x,y) = p_t(y-x) = (2\pi t)^{-d/2} \exp\Big(-\frac{|y-x|^2}{2t}\Big), \qquad x,y \in \mathbb{R}^d,\ t>0.$$

$$G(x,y) = G(y-x) = \int_0^\infty p_t(x,y)\,dt = \gamma_d |y-x|^{2-d}, \qquad x,y \in \mathbb{R}^d,\ d \ge 3.$$

The letters C, c, c_1, c_2 etc. are often used to denote positive constants whose exact value is not specified.